

Question 1. Sketch the domain of $f(x, y)$.

(1) $f(x, y) = \frac{1}{\ln(y - x - 2)}$

(2) $f(x, y) = \sqrt{4 - x^2 - y^2}$

(3) $f(x, y) = \sqrt{(x-1)(y-2)}$

(4) $f(x, y) = \sec(x)\tan(y)$

Question 2. Sketch the level curves of $f(x, y)$.

(1) $f(x, y) = \sqrt{x^2 + y^2}$

(2) $f(x, y) = 4x^2 - y^2$

(3) $f(x, y)$ is the distance of the point (x, y) from the line $y - 2x + 2 = 0$.

(4) (challenge) $f(x, y)$ is the angle ABC formed by $A = (-1, 0)$, $B = (x, y)$, and $C = (1, 0)$ for $y > 0$.

Question 3. Evaluate the following limits or show that the limit does not exist.

$$(1) \lim_{(x,y) \rightarrow (0,\pi/4)} \sec(x) \tan(y)$$

$$(2) \lim_{\substack{(x,y) \rightarrow (1,1) \\ x \neq y}} \frac{x+y-2}{x-y}$$

$$(3) \lim_{(x,y) \rightarrow (1,1)} \frac{x-y}{x^2-y^2}$$

$$(4) \lim_{(x,y) \rightarrow (0,0)} \frac{2xy}{x^2+y^2}$$

$$(5) \lim_{(x,y) \rightarrow (0,0)} x \cos \frac{1}{y}$$

$$(6) \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2 + y^2}$$

$$(7) \lim_{(x,y) \rightarrow (0,0)} \frac{\sin(x^2 + 2y^2)}{x^2 + y^2}$$

$$(8) \lim_{(x,y) \rightarrow (0,0)} e^y \frac{\sin x}{x}$$